Task 2: 10-Word Speech Recognition under Noise

Oct 17, 2026 · @Gedas

Abstract

A small CNN was trained on log-mel spectrograms of 4,002 clean recordings of 10 words and tested on 998 recordings from other speakers. It recognises 77.3 % of clean test words; with added noise accuracy drops to 29–40 % at 10 dB SNR and close to chance at 0 dB. White noise is more harmful than real noise, and under strong noise the model answers "yes" for most recordings.

1. Dataset preparation

The experiment uses the **Google Speech Commands Dataset v0.02** (P. Warden, "Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition", arXiv:1804.03209, 2018): one-second recordings of single English words by thousands of volunteers, mono WAV at **16 kHz**. Selected words: *yes, no, up, down, left, right, on, off, stop, go*. To keep the experiment small, **500 recordings of each word were chosen at random**, giving a balanced set of **5,000 recordings from 1,795 speakers**.

Training and test sets. The split is speaker-independent. The speaker is the part of the file name before `_nohash_` (e.g. `004ae714_nohash_0.wav` is speaker `004ae714`). The speakers were shuffled and 20 % of them (359) were put in the test set; the recordings of the other 1,436 speakers form the training set. Whole speakers go to one set, so no speaker is in both (the program prints "Speakers in both sets: 0"), and all 10 words are in both sets (Table 1).

Set	yes	no	up	down	left	right	on	off	stop	go	Total	Speakers
Training	388	397	401	389	408	405	412	411	393	398	4,002	1,436
Test	112	103	99	111	92	95	88	89	107	102	998	359

Table 1. Number of recordings per word in the training and test sets.

2. Feature extraction

Every recording is converted into a **log-mel spectrogram**, which shows the energy at each frequency over time. Words differ in this pattern: vowels have strong energy bands (formants), and /s/ has noise-like energy at high frequencies. The **mel scale** gives more detail at low frequencies, like human hearing, and the **logarithm** (dB) makes quiet and loud parts comparable. The result is a small 2-D "image" of the word, a good input for a CNN.

Preprocessing. Recordings were loaded at 16 kHz with `librosa` and padded with zeros to exactly 1 s (16,000 samples), because a CNN needs inputs of the same size. Features were computed with the same parameters for all recordings (Table 2) and normalised with the training-set mean and standard deviation.

Parameter	Value	Parameter	Value
Sampling rate	16 kHz	FFT size	512
Window length	400 (25 ms)	Mel filters	40
Hop length	160 (10 ms)	Scale	dB
Window	Hann	Feature size	40 × 101

Table 2. Feature extraction parameters.

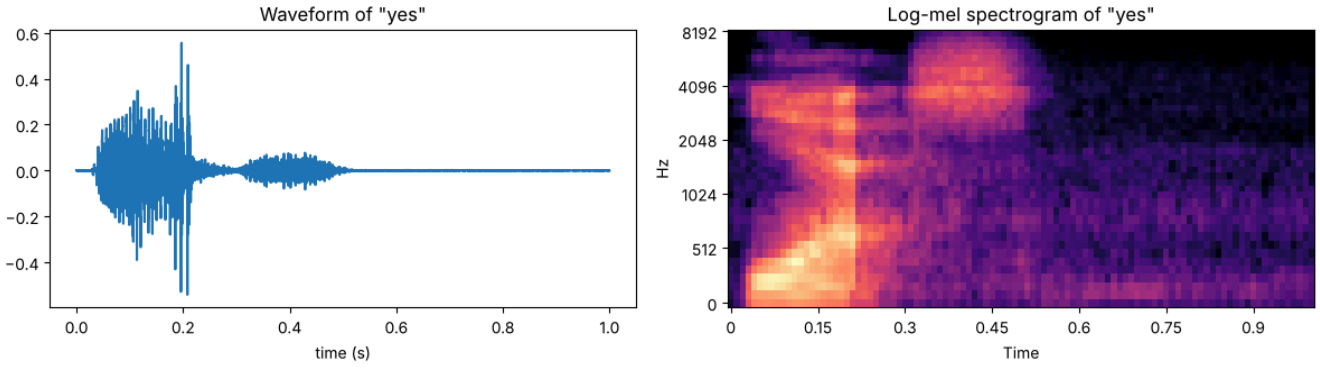


Figure 1. Waveform (left) and log-mel spectrogram (right) of one recording of "yes". Brighter colour means more energy.

Interpretation. The word "yes" has two parts. From 0.03 to 0.22 s comes the voiced part /jɛ/: a large waveform and strong energy below about 1 kHz, with bright bands that are the vowel's formants. From 0.3 to 0.5 s comes the /s/: a small waveform, but energy concentrated at high frequencies (about 3–6 kHz). After 0.55 s the recording is silent and the spectrogram is dark.

3. Model

A simple **CNN** built with PyTorch was used for all experiments. Its convolutional layers learn small time–frequency patterns (such as a formant or an /s/), max-pooling shrinks the image, and the fully connected layers decide which word was spoken.

#	Layer	Details	Output size
1	Conv + ReLU + max-pool	16 filters 3×3, pool 2×2	16 × 20 × 50
2	Conv + ReLU + max-pool	32 filters 3×3, pool 2×2	32 × 10 × 25
3	Flatten		8,000
4	Fully connected + ReLU	64 neurons, dropout 30 %	64
5	Fully connected (output)	10 neurons (softmax)	10

Table 3. CNN architecture (input 1 × 40 × 101; 517,514 parameters).

Training. Only on the clean training set (4,002 recordings, 1,436 speakers), with **cross-entropy loss**, **Adam** (learning rate 0.001), **15 epochs** and **batch size 32**. The average loss fell from 2.08 to 0.45.

4. Testing under noise

Procedure. The model was tested on the same 998 test recordings in three conditions, with noise added only to the test audio: (1) **clean speech**; (2) **white Gaussian noise** (`np.random.randn`); (3) **real background noise**: a random 1-s segment of one of four dataset recordings (*doing the dishes, cat miaowing, exercise bike, running tap*), chosen at random per recording. The noise was scaled to SNR = $10 \cdot \log_{10}(P_signal / P_noise)$, with P the average power over the clip, at 30, 20, 10 and 0 dB.

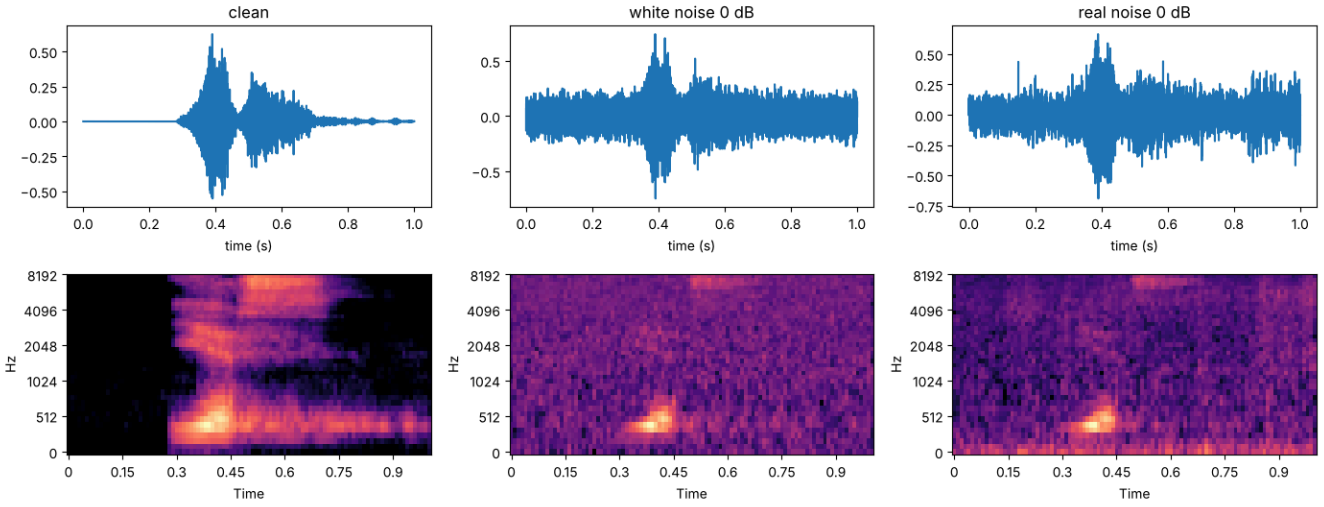


Figure 2. A test recording of "yes": clean, with white noise and with real noise at 0 dB.

In Figure 2, white noise fills the empty areas around the word at all frequencies; only the loudest part of the vowel (0.35–0.45 s, below 1 kHz) stays clear and the /s/ is hidden. The real noise is less even, so more of the word remains visible.

Accuracy. Table 4 and Figure 3 show the accuracy in every condition (chance level for 10 words: 10 %).

Noise	Clean	30 dB	20 dB	10 dB	0 dB
White Gaussian noise	77.3 %	69.6 %	56.1 %	29.3 %	14.2 %
Real background noise	77.3 %	71.7 %	62.4 %	40.1 %	19.3 %

Table 4. Test accuracy for clean speech and for each noise type and SNR.

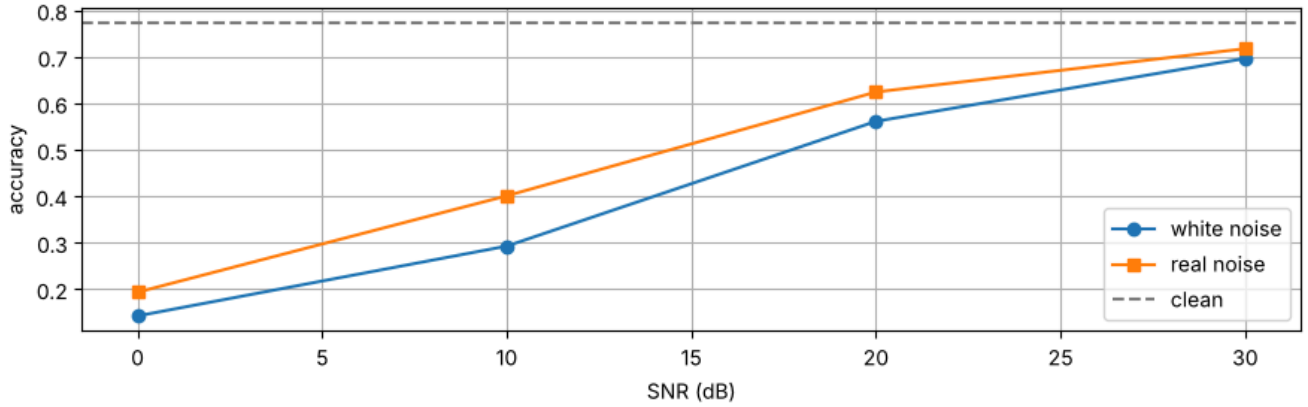


Figure 3. Test accuracy versus SNR; the dashed line is the clean accuracy.

Effect of SNR. Accuracy falls at every step: weak noise (30 dB) already costs 6–8 points, at 10 dB less than half of the words are recognised, and at 0 dB accuracy is near chance. The model never saw noise in training.

Effect of noise type. Real noise is less harmful than white noise at every SNR (by 2.1, 6.3, 10.8 and 5.1 points). White noise covers all frequencies at all times; real noise is concentrated in some bands or comes in bursts, so parts of the word stay visible.

Word	Clean	White 10 dB	Real 10 dB	White 0 dB	Real 0 dB
yes	0.89	0.96	0.85	0.93	0.85
no	0.70	0.13	0.28	0.00	0.01
up	0.81	0.09	0.25	0.00	0.08
down	0.65	0.03	0.12	0.00	0.00
left	0.77	0.10	0.33	0.00	0.11
right	0.84	0.11	0.39	0.00	0.14

Word	Clean	White 10 dB	Real 10 dB	White 0 dB	Real 0 dB
on	0.84	0.01	0.12	0.00	0.01
off	0.75	0.66	0.66	0.04	0.25
stop	0.79	0.56	0.60	0.32	0.34
go	0.70	0.20	0.36	0.00	0.07

Table 5. Accuracy of each word (share of its test recordings recognised correctly).

Effect of word identity. On clean speech the words score 0.65 (*down*) to 0.89 (*yes*). Under noise *yes, stop* and *off* stay high, while *no, down, on* and *up* drop to almost zero. The robust words contain a noise-like fricative (/s/ in *yes* and *stop*, /f/ in *off*); added noise looks like a fricative, so the model prefers these words, above all *yes*. Their high accuracy is a bias, not real robustness.

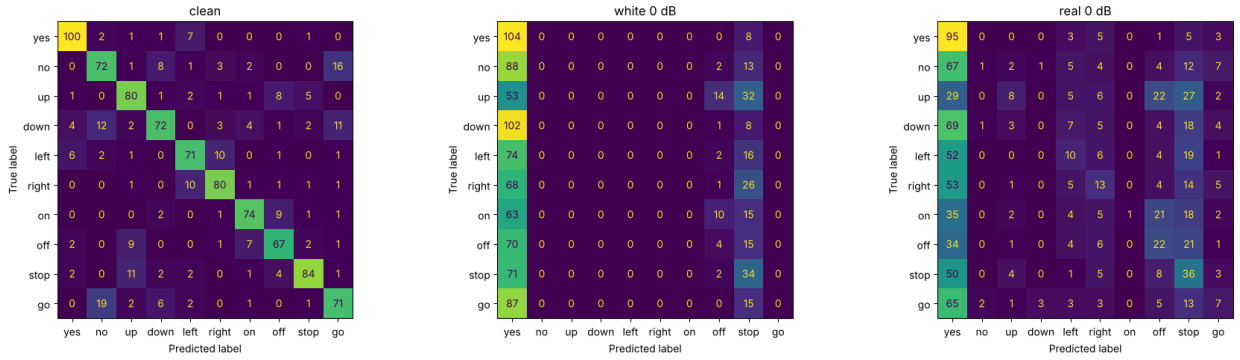


Figure 4. Confusion matrices: clean, white noise 0 dB, real noise 0 dB (rows: true words).

Confusions. On clean speech (Figure 4, left) the main mistakes are between similar-sounding words: *go* → *no* (19) and *no* → *go* (16), sharing the vowel /oʊ/; *down* → *no* (12) and *down* → *go* (11); *stop* → *up* (11), sharing the final /p/; *left* ↔ *right* (10 each). Under noise almost everything is classified as *yes*: with white noise at 0 dB, 780 of 998 recordings (78 %), with the largest confusions *down* → *yes* (102), *no* → *yes* (88) and *go* → *yes* (87); with real noise at 0 dB, 549 (55 %).

5. Conclusions

1. **SNR:** from 77.3 % on clean speech, accuracy falls with every step of noise, to 30–40 % at 10 dB and near chance at 0 dB.
2. **Noise type:** white noise is always more harmful than real noise at the same SNR.
3. **Words:** noise biases the model toward fricative words, above all *yes*, replacing the clean confusions between similar words (*go* ↔ *no, left* ↔ *right*).
4. Noisy training data would likely make the model more robust, but was outside the task's scope.